

Non-commutative crystalline cohomology

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Crystalline cohomology

k perfect field of char $p > 2$

$W_m(k)$ Witt vectors $(k = \mathbb{F}_p, W_m(k) = \mathbb{Z}/p^m \mathbb{Z})$

$$R\Gamma_{\text{cris}} : S_m/k \longrightarrow D(W_m(k)\text{-mod})$$

$$\textcircled{1} \quad R\Gamma_{\text{cris}} \bigotimes_{W_m(k)}^{\mathbb{L}} k = R\Gamma_{\text{DR}}$$

$$\textcircled{2} \quad \tilde{X} \in S_m/W_m(k), \quad \tilde{X} \otimes_{W_m(k)}^{\mathbb{L}} k \simeq X,$$

$$R\Gamma_{\text{cris}}(X) = R\Gamma_{\text{DR}}(\tilde{X}/W_m(k))$$

$$R\Gamma_{\text{cris}, m} \otimes_{W_m(k)}^{\mathbb{L}} W_{m-1}(k) = R\Gamma_{\text{cris}, m-1}$$

Cor $\tilde{X} \in S_m/W_m(k)$, $\tilde{\varphi} \sim \tilde{X}$, $\tilde{\varphi} \equiv \text{Id} \pmod{p}$. Then

$$\text{id} = \tilde{\varphi}^* \sim R\Gamma_{\text{DR}}(\tilde{X}).$$

(Cartan's formula.

$$Y/S \text{ smooth}, \theta', \theta \in \Gamma(Y, T_{Y/S}), \quad \Omega_{Y/S}^{\bullet} \xrightarrow{\iota_{\theta}} \Omega_{Y/S}^{\bullet-1}$$

$$[\iota_{\theta}, d_{\text{DR}}] = L_{\theta}$$

$$[L_{\theta}, \iota_{\theta'}] = \iota_{[\theta, \theta']}, \quad [L_{\theta}, L_{\theta'}] = 0$$

pt of ω $\tilde{\varphi}^* \mapsto \partial \tilde{x}$ $\tilde{\varphi}^* \equiv \text{Id} \mod p$

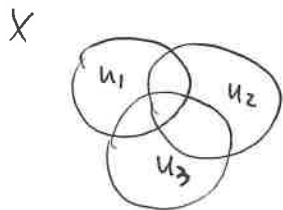
$$\Gamma(\tilde{x}, T_{\tilde{x}}/\omega_m) \ni \theta = \log \tilde{\varphi}^* = - \sum_n \frac{(1 - \tilde{\varphi}^*)^n}{n}, \quad \theta \equiv 0 \mod p$$

$$\Omega_{\tilde{x}}^* \ni \tilde{\varphi}^* = \sum \frac{L_{\theta}^n}{n!}$$

$$h_{\varphi} = \sum \frac{L_{\theta} L_{\theta}^{n-1}}{n!}, \quad [d_{\text{DR}}, h_{\varphi}] = \tilde{\varphi}^* - 1$$

$$h_{\tilde{\varphi} \circ \tilde{\varphi}} = h_{\tilde{\varphi}} + \tilde{\varphi} \circ h_{\tilde{\varphi}}$$

$$G \times G \times G \rightrightarrows G \times G \rightrightarrows G$$



$$\tilde{u}_i / \omega_m(k)$$

$$\bigoplus_i \Omega_{\tilde{u}_i / \omega_m(k)} \rightarrow \bigoplus_{i,j} \Omega_{\tilde{u}_i \wedge \tilde{u}_j} \rightarrow$$

A/R

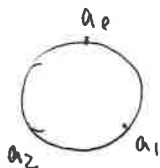
smooth commutative

$$\begin{array}{ccc} \Omega_{A/R}^n & = & \text{Tor}_n^{A \otimes A^{\text{op}}} (A, A) = HH_n(A) \\ \downarrow d_{\text{DR}} & & \downarrow B \leftarrow \text{defined for any assoc. alg.} \\ \Omega_{A/R}^{n+1} & = & \text{Tor}_{n+1}^{A \otimes A^{\text{op}}} (A, A) \end{array}$$

$$\begin{array}{c} \hookleftarrow \\ \rightarrow A \otimes A \otimes A \xrightarrow[\delta_H]{B} A \otimes A \xrightarrow[\delta_H]{B} A : C(A) \\ a_0 \otimes a_1 \mapsto a_0 a_1 - a_1 a_0 \end{array}$$

$$- B^2 = 0$$

$$- [B, \delta_H] = 0$$



$$a_0 \otimes a_1 \otimes a_2 \mapsto a_0 a_1 \otimes a_2 - a_0 \otimes a_1 a_2 + a_2 a_0 \otimes a_1$$

$$\deg u = 2.$$

$$\left(C(A)(u), \delta_H + B u \right) =: CP(A)$$

$$A/R \quad \text{assoc. flat over } R$$

$$\theta \in \text{Der } A/R, \quad L_\theta \simeq CP(A/R) \supset L_\theta$$

$$[\delta, L_\theta] = L_\theta$$

$$[L_\theta, L_{\theta'}] = L_{[\theta, \theta']}$$

$$1+\theta \simeq A[\epsilon]/R[\epsilon]$$

$$1+\theta \simeq CP(A[\epsilon]/R[\epsilon])$$

$$[L_\theta, L_{\theta'}] \neq 0$$

$$\text{"id"} \in L_\theta$$

$$\text{char } k = p > 2, \quad k \text{ perfect}$$

$$CP^{cris} : \left\{ \begin{array}{c} \text{algebras} \\ \text{over } k \end{array} \right\} \longrightarrow D(W_m(k)\text{-mod})$$

$$\textcircled{1} \quad CP^{cris} \otimes_{W_m(k)} k = CP$$

$$\textcircled{2} \quad \tilde{A}/W_m(k) \text{ flat, } \tilde{A} \otimes_{W_m(k)} k = A,$$

$$CP^{cris}(A) \simeq CP(\tilde{A})$$

$$\dots \rightarrow A^{i-1} \xrightarrow{d} A^i \xrightarrow{d} A^{i+1} \xrightarrow{d} \dots$$

$$, d \in \text{Der}(A), \quad \deg d = 1.$$

$$[d, d] = 0$$

$$\left(CP\left(\bigoplus_i A^i\right), \delta + L_d \right) =: CP(A/R)$$

$$\left(\bigoplus_i A^i\right) \otimes \left(\bigoplus_j A^j\right)$$

$$\tilde{A}, \tilde{a} \in \text{Der}(\tilde{A})$$

$$\frac{1}{2}[\tilde{a}, \tilde{a}] = \tilde{a}^2 \neq 0$$

$$\text{CP}(\oplus \tilde{A}_r), \quad s + L\tilde{a} + L\tilde{a}^2 + \dots$$

$$\tilde{A}/k$$

$$\text{CP}^{\text{crs}}(\tilde{A}) = \text{CP}(\tilde{A}/W_m(k)) \otimes_{\text{CP}(k/W_m(k))} W_m(k)$$

$$\mathbb{F}_p$$

$$\text{CP}\left(\mathbb{Z}_p \xrightarrow{f} \mathbb{Z}_p/\mathbb{Z}_p\right) = \text{CP}\left(\mathbb{Z}_p \xrightarrow{0} \mathbb{Z}_p/\mathbb{Z}_p\right)$$